Seminar 3

Inside a bank: fragile by design

Complements Lecture 3: what a bank does, why it breaks, and pricing the risk

The big picture: why fragility matters

A bank is the most leveraged business you will ever study. It funds itself almost entirely with other people's money, which makes it extraordinarily useful, and extraordinarily easy to break.

1. A run at the speed of a tap

In March 2023, Silicon Valley Bank saw about $42bn of deposits demanded in a SINGLE day, with more queued for the next.

It was a sound-looking bank on Wednesday and gone by Friday. Mobile banking made the run instant.

2. A wafer-thin cushion

A typical bank holds only about £5 of equity for every £100 of assets.

So a fall of just 5% in the value of its loans can wipe the owners out completely. Compare a shop at 25% or Apple at about 20%.

Banks run on trust and a tiny cushion. Today you'll compute both the losses they expect and the ones that break them.

Where this fits, and today's aim

From Lecture 3 (the ideas)

- A bank turns short, safe deposits into long, risky loans (transformation), and creates most of the money supply.
- It is fragile: about 5% equity, deposits repayable on demand.
- A safety net (capital, deposit insurance, lender of last resort) sits underneath.

This seminar (what you'll DO)

- price the credit risk of a loan with expected loss = PD x LGD x amount;
- show how a 5% equity cushion is wiped out by a small fall in asset values;
- decide which failing bank to rescue: the illiquid one or the insolvent one.

A bank is leverage at the extreme

Remember Seminar 2: debt amplifies risk. A bank takes that to the limit.

A supermarket	Apple	A bank
~25% equity	~20% equity	~5% equity

The bank's other 95% is depositors' and lenders' money. The owners' stake, the cushion against losses, is tiny.

Two consequences follow, and we compute both today: (1) losses it EXPECTS on its loans, priced into the interest rate; (2) UNEXPECTED losses, which eat straight into that 5% cushion.

Pricing credit risk: expected loss

Expected loss = PD x LGD x amount

- PD, probability of default: the chance this type of borrower fails.
- LGD, loss given default: the share of the loan the bank loses if they do.

Worked: a £2,000 loan. This type of borrower defaults 4% of the time (PD), and the bank loses 25% when they do (LGD).

Expected loss = 4% x 25% x £2,000 = £20.

So the bank builds £20 into the interest rate it charges: expected losses are paid out of revenue, not the cushion.

Activity: expected vs unexpected loss

Your turn: a £10,000 loan with a 5% probability of default and a 50% loss given default. What is the expected loss? Then: what happens in a bad year when defaults are far worse than expected?

Expected loss

5% x 50% x £10,000 = £250

Charged for in the interest rate, met from revenue.

Unexpected loss

If losses come in at £900, the extra £650 was NOT priced in.

That surplus loss eats into EQUITY, the 5% cushion.

Expected losses come out of revenue; unexpected losses come out of equity. Understanding that one line is the key to understanding banks.

Activity: how much can the cushion take?

Northshore Bank has £200 of loans, funded by £190 of deposits and £10 of equity (a 5% cushion). Work out two things on your own.

Q1: the loans fall by £6

Assets: 200 to 194.

Equity: 10 to 4.

Still covers the £190 deposits, so solvent, but only just.

Q2: what loss wipes equity out?

Any loss above £10, a fall of just 5%.

At -£10 the assets (190) exactly equal deposits; beyond that, depositors lose.

A bank with a 5% cushion is insolvent after just a 5% fall in its assets. That is what 'fragile' means, precisely.

Activity: diagnose, then decide

Two banks fail this morning. From the numbers, work out which is ILLIQUID (sound but stuck for cash) and which is INSOLVENT (owes more than it owns). Which do you rescue?

Riverbank

Loans worth £104, deposits £100. But £30 is demanded today and it holds only £2 in cash.

Diagnosis: solvent (104 > 100) but ILLIQUID: it cannot find £30 today.

Peakwater

Loans have gone bad and are now worth only £96, against deposits of £100.

Diagnosis: INSOLVENT: it owes £4 more than it owns. A loan only delays the end.

Rescue Riverbank: the central bank lends to the solvent-but-illiquid against its good loans. Do not prop up the insolvent Peakwater.

In your groups: let it fail?

A mid-size bank is wobbling this week. As a group, decide: let it fail, guarantee its deposits, or arrange a rescue by a bigger bank? And what must you find out first?

What you need to know first

Is it illiquid (sound but short of cash today) or insolvent (owes more than it owns)? How many ordinary depositors are exposed? Could its failure spread to other banks?

The dilemma: moral hazard

Rescue every bank and they take reckless risks next time, knowing they will be saved. Rescue none and a healthy-but-illiquid bank can be killed by pure panic.

Diagnose first (illiquid or insolvent), then weigh contagion against moral hazard. This is the exact judgement regulators faced in 2008.

Three traps to avoid in the exam

1

Confusing expected and unexpected loss

Expected loss (PD x LGD x amount) is met from REVENUE. It is the UNEXPECTED loss that hits equity.

2

Treating illiquid and insolvent as the same

Illiquid = sound but no cash today (a loan helps). Insolvent = owes more than it owns (a loan won't help).

3

Forgetting equity absorbs losses first

Depositors only lose once the equity cushion is gone. With about 5% equity, that does not take much.

Exam-style question

A £5,000 loan has a 2% probability of default and a 40% loss given default. The expected loss is:

A) £40

B) £100

C) £400

D) £2,000

Answer: A. Expected loss = 2% x 40% x £5,000 = £40.

Take these away

- A bank runs on about 5% equity, so a small fall in its assets can wipe the owners out.
- Expected loss = PD x LGD x amount, and it is charged for in the interest rate (paid from revenue).
- Unexpected losses are what hit the equity cushion.
- Illiquid (sound but no cash today) is not insolvent (owes more than it owns). Only the first should be rescued.

Before Seminar 4: be able to compute an expected loss and explain in one line why a bank is illiquid but not insolvent, or the reverse.